

Learning by Teaching A Self-representation Agent: A Self-Guided Learning Framework for Human-Teachable Agent Interaction

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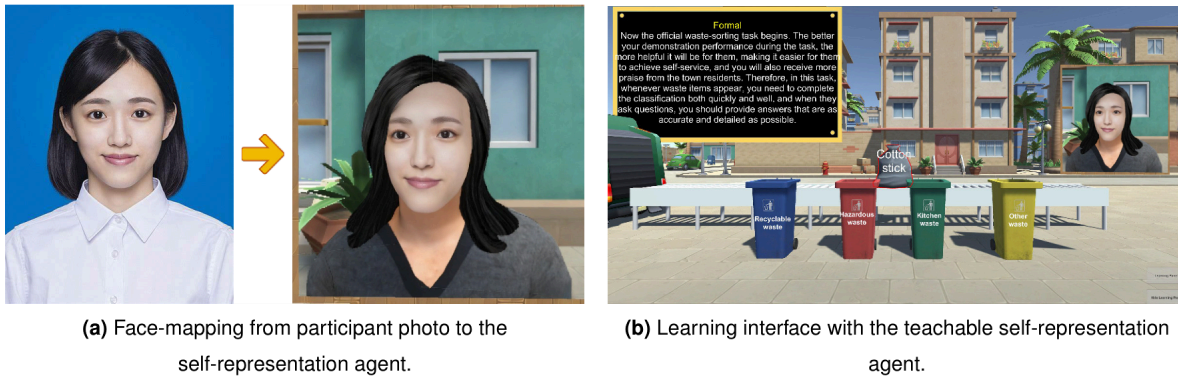


Figure 1: The figure shows a Learning Framework for Human-Teachable Agent Interaction which support Human users to learn by teaching a self-representation agent.

ABSTRACT

In virtual learning environments (VLEs), Learning from Teaching a Teachable Agent (LFT) has demonstrated notable benefits for enhancing learning motivation and experience. To advance this paradigm, we focused on the role of self-schema in learning and proposed a novel framework: Learning from Teaching a Self-representation Agent (LFTS). To empirically validate the effectiveness of LFTS, we developed a prototype system and conducted a 2 (learning mode: teaching vs. being taught) $\times$ 2 (agent mode: self-representation agent vs. other-representation agent) within-subjects experiment. Results showed that LFTS significantly improved agent perception, affective appraisal, flow experience, and fostered stronger human-agent relationships in VLEs.

Index Terms: Learning by teaching; Teachable agents; Virtual learning environments; Intelligent agents; Flow experience

1 INTRODUCTION

Virtual learning environments (VLEs) increasingly incorporate animated pedagogical agents (APAs) to provide guidance, feedback, and socio-emotional support. Prior work suggests that such agents can improve learners' motivation and experience, an effect often described as the *Persona Effect* [6, 8]. However, these benefits are not consistently observed: when the agent's identity and interaction

style are weakly aligned with the learner and the task, both experience and outcomes can be diminished [1, 4].

A major reason is that most APA designs still treat the agent as an "other". Existing studies have largely explored surface-level factors—appearance [5], personality and socio-emotional traits [2], and role positioning such as teacher/peer/student [6]—but have paid comparatively less attention to *self-relevance*. Yet psychological evidence indicates that self-referential processing facilitates deeper engagement when information relates to one's self-schema [7, 9, 10], and Self-Determination Theory further suggests that sustained motivation depends on satisfying autonomy, competence, and relatedness needs [3]. Together, these perspectives motivate agent designs that resonate with learners' identity rather than merely providing generic instruction. One representative approach that partially addresses learner involvement is the use of *teachable agents*, which are interactive agents designed to acquire knowledge by being taught by learners, thereby positioning learners in the role of a teacher rather than a passive recipient [2]. While teachable agents leverage the learning-by-teaching effect to promote active explanation and reflection, they typically remain generic "others" and do not explicitly incorporate the learner's own identity or self-schema.

To address this gap, we propose **Learning from Teaching a Self-representation Agent (LFTS)**. In LFTS, learners assume the role of a teacher and instruct a *self-schema* teachable agent whose appearance and identity cues are modeled after the learner. The agent's knowledge is not autonomously improved; instead, it is constructed and updated through the learner's teaching, turning the interaction into a form of "teaching the self." We implement LFTS in a VLE prototype (Fig. 1) and evaluate it in a within-subjects study that manipulates learning mode (teaching vs. be-

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ing taught) and agent identity (self-representation agent vs. other-representation agent). Our results show that self-representation teaching can enhance learners' flow and affective experience, and strengthen human-agent relationship quality.

2 SYSTEM DESIGN

To instantiate the proposed LFTS paradigm, we designed an immersive VLE system that integrates self-representation with teaching-based interaction, enabling learners to acquire knowledge by teaching a self-representation agent. Fig. 1(a) shows the face-mapping process, and Fig. 1(b) shows the learning interface.

2.1 Design Rationale

Learning-by-teaching encourages learners to organize, articulate, and reflect on their knowledge, promoting engagement and deeper processing. Self-representation increases identity relevance and emotional involvement by aligning the agent with the learner's self-concept. Together, LFTS reframes teaching from instructing an external agent into a reflective process of "teaching oneself."

2.2 System Overview

We developed a lightweight virtual learning system to instantiate the proposed LFTS paradigm in a virtual learning environment. In LFTS, learners assume the role of a teacher and instruct a teachable agent with *self-representation* cues modeled after the learner (e.g., via face/feature mapping), reframing the interaction as a reflective process of "teaching the self." The system supports an identity-aligned interaction cycle in which learners explain domain knowledge and examples to the agent, the agent actively asks clarification questions and requests guidance, and the taught content is immediately grounded through task execution and feedback, enabling an initial empirical examination of how self-representation agent design shapes experiential and relational outcomes.

3 SYSTEM IMPLEMENTATION

We implemented a functional prototype that supports real-time interaction with a teachable agent in a VLE while preserving the learner-aligned *self-representation* throughout the session. Before each learning session, the system generates a self-representative agent by mapping the learner's facial features onto an avatar and maintains this identity-aligned appearance during the interaction to support self-referential engagement. Crucially, the agent is explicitly designed as a novice learner whose responses and behaviors are driven by the learner's instruction, ensuring that learning emerges from the act of teaching rather than agent-driven tutoring. This implementation is sufficient for evaluating the LFTS paradigm in a controlled user study.

4 USER STUDY

We conducted a user study to examine how teaching a self-representation agent shapes learning experience and human-agent relationship outcomes rather than maximizing task performance.

Thirty-two participants (aged 18–40) completed a 2 (learning mode: teaching vs. being taught) $\times$ 2 (agent identity: self-representation vs. other-representation) within-subjects experiment with four counterbalanced conditions.

Participants used the prototype in a virtual garbage classification task: in the teaching condition, they explained rules and examples to the agent; in the being-taught condition, they received guidance. The agent appeared either as a self-representation avatar or an unrelated other-representation agent, and participants reported their subjective experience after each condition. This study adhered to the relevant guidelines of the Declaration of Helsinki and was approved by the Human Research Ethics Committee of institute. For all procedures involving human participants, informed consent was obtained from each participant.

We measured flow experience and perceived human-agent relationship quality. Results showed a significant effect of agent identity on experiential outcomes: overall flow was higher with self-representation agents than with other-representation agents, $F_{(1,31)} = 8.745, p = 0.006$, with the same trend observed on engagement and fluency factors ($ps < 0.01$).

In terms of involvement, a significant interaction was observed between learning mode and agent identity, $F_{(1,31)} = 6.81, p = 0.014, \eta_p^2 = 0.180$. Simple effects analyses showed that under the teaching mode, involvement was significantly higher when participants interacted with a self-representation agent than with an other-representation agent ($M\text{-diff} = 1.219, p < 0.001, \eta_p^2 = 0.406$). In contrast, no significant difference between agent identities was observed in the being-taught mode.

Self-representation agents were also associated with stronger human-agent relationships. Participants reported significantly higher perceived closeness and greater intimacy when interacting with self-representation agents ($ps < 0.01$).

5 CONCLUSION

In summary, this work contributes a LFTS paradigm that integrates self-representation agent with teaching-based interaction in virtual learning environments and reframes learning-by-teaching as a reflective process of "teaching the self". Through a user study, we found that teachable self-representation agents can enhance learners' flow experience and perceived human-agent relationship, providing initial evidence that identity alignment can enhance learners' experience and perceived human-agent connection.

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